

```
(%i3) /* Symbols */
declare([x, x_k, h], real)$;
assume(h > 0)$;
depends(f, x)$;
```

```
(%i4) /* Expanding f(x) at the midpoint between x_k and x_{k+1} */
series: taylor(f(x), x, x_k + h / 2, 10);
```

series

$$f\left(\frac{2x_k + h}{2}\right) + \left(\frac{d}{dx} f(x) \Big|_{x=\frac{2x_k + h}{2}}\right) \left(-x_k + x - \frac{h}{2}\right) + \frac{\left(\frac{d^2}{dx^2} f(x) \Big|_{x=\frac{2x_k + h}{2}}\right) \left(-x_k + x - \frac{h}{2}\right)^2}{2} +$$

$$\frac{\left(\frac{d^3}{dx^3} f(x) \Big|_{x=\frac{2x_k + h}{2}}\right) \left(-x_k + x - \frac{h}{2}\right)^3}{6} + \frac{\left(\frac{d^4}{dx^4} f(x) \Big|_{x=\frac{2x_k + h}{2}}\right) \left(-x_k + x - \frac{h}{2}\right)^4}{24} +$$

$$\frac{\left(\frac{d^5}{dx^5} f(x) \Big|_{x=\frac{2x_k + h}{2}}\right) \left(-x_k + x - \frac{h}{2}\right)^5}{120} + \frac{\left(\frac{d^6}{dx^6} f(x) \Big|_{x=\frac{2x_k + h}{2}}\right) \left(-x_k + x - \frac{h}{2}\right)^6}{720} +$$

$$\frac{\left(\frac{d^7}{dx^7} f(x) \Big|_{x=\frac{2x_k + h}{2}}\right) \left(-x_k + x - \frac{h}{2}\right)^7}{5040} + \frac{\left(\frac{d^8}{dx^8} f(x) \Big|_{x=\frac{2x_k + h}{2}}\right) \left(-x_k + x - \frac{h}{2}\right)^8}{40320} +$$

$$\frac{\left(\frac{d^9}{dx^9} f(x) \Big|_{x=\frac{2x_k + h}{2}}\right) \left(-x_k + x - \frac{h}{2}\right)^9}{362880} + \frac{\left(\frac{d^{10}}{dx^{10}} f(x) \Big|_{x=\frac{2x_k + h}{2}}\right) \left(-x_k + x - \frac{h}{2}\right)^{10}}{3628800} + \dots$$

```
(%i5) /* Sum over k to compute the full integral of f(x) from a to b */
expand(integrate(series, x, x_k, x_k + h));
```

(%o5)

$$h f\left(x_k + \frac{h}{2}\right) + \frac{h^{11} \left(\frac{d^{10}}{dx^{10}} f(x) \Big|_{x=x_k + \frac{h}{2}}\right)}{40874803200} + \frac{h^9 \left(\frac{d^8}{dx^8} f(x) \Big|_{x=x_k + \frac{h}{2}}\right)}{92897280} + \frac{h^7 \left(\frac{d^6}{dx^6} f(x) \Big|_{x=x_k + \frac{h}{2}}\right)}{322560} +$$

$$\frac{h^5 \left(\frac{d^4}{dx^4} f(x) \Big|_{x=x_k + \frac{h}{2}}\right)}{1920} + \frac{h^3 \left(\frac{d^2}{dx^2} f(x) \Big|_{x=x_k + \frac{h}{2}}\right)}{24}$$